



The Bruhat–Tits Tree Code: Status and Test Plan

Rowan Brad Quni-Gudziñas · QNFO — CWI Summer School, 26 Aug 2026

1 · The code in one line

Holographic **perfect-tensor codes on a truncated Bruhat–Tits tree**: a $[[3,1,1]]$ perfect tensor sits at every internal vertex, logical information lives at the root, physical qubits at the leaves, and error correction is built into the tree geometry rather than applied by active measurement loops.

Error model	Tree code	Surface code	Ratio
Bit-flip	50.0%	~10.9%	4.6×
Depolarizing	75.0%	~1.0%	75×
Independent X+Z	17.30%	—	—

Computed three independent ways — analytical recursion, exact Gottesman–Knill stabilizer simulation, Monte Carlo — with the verification script (35 checks) deposited with the guide record.

All values are classical simulation: i.i.d. noise, crossing criterion, no hardware. Surface comparators are the standard published figures (bit-flip ~10.9%; depolarizing ~1% circuit-level).

2 · Status (simulation only — no hardware yet)

Verified computationally, three ways. Thresholds from analytical recursion, exact stabilizer simulation, and Monte Carlo; the verification script (35 checks) is deposited with the guide record (10.5281/zenodo.22038733).

Published boundary. Under independent errors the related qudit family sits at $\approx 2 \times 10^{-4}$, about 55× below surface codes. The advantage claim is scoped to *correlated-failure* regimes — which is exactly what test T1 probes.

An external no-go was engaged. The QEC–Darwinism tradeoff ($F_L > 0.874$) was audited on tree-code spaces: collective refinement moves the first-redundancy boundary to $F_L \approx 0.83$ –0.85, with a disclosed residual gap.

Self-correction published. A proposed p-adic reading of stabilizer codes turned out to carry no new content; the correction is published rather than the confirmation.

What is missing: hardware data. All numbers above are classical simulation; nothing has run on a quantum processor — tests T3–T4 are the instrumentation path.

3 · Test plan (near-term first)

#	Test	Where	Disconfirmation criterion
T1	Matched-noise threshold comparison: tree vs surface under the <i>same</i> correlated noise model	Classical	Tree advantage falls to parity under matched noise
T2	Decoder complexity + full overhead (physical qubits + classical cost)	Classical	Advantage disappears once decoding cost is included
T3	Return-probability decay of a random walk on a p-regular tree graph	Analog / ion sim, ~8 wk	No change in decay exponent at the predicted value
T4	Clock-rest ultrametricity split: diagonal 0% vs non-diagonal 29–35%	Trapped ion, ~8 wk	Measured split outside the predicted range
T5	Independent re-implementation of T1 by an external group	Community	Reproducibility test, not a prediction

T1 and T2 are pure classical work — cheap, decisive, and doable immediately; they answer the two objections this room will raise first. Every criterion is numeric, so a null result is publishable.

4 · What we ask of this room

1. An openly specified **correlated-noise benchmark** (data + model) we can run T1 against.
2. Decoder reality check: does a known decoder family already cover tree-structured codes, and how does its complexity scale?
3. Pointers on **perfect tensors for $p > 2$** (binary $[[3,1,1]]$ is settled; larger primes are open).
4. **Independent re-run of T1** — we hand over the simulator and the protocol; a third-party rerun settles the classical claim.

Horizons

2026: T1–T2 + open simulator release · **2026–27:** T3–T4 instrumentation, noise-model publication · **Long term:** prototype only if the matched-noise scrutiny survives; energy accounting (joules per correct answer) as the comparison.